from Beginner to Master

DevOps를 위한 Docker 가상화

Section 9. Advanced Docker Usage

프로필

Dowon Lee

수강생 19K 강의평점 4.8(1K)

멘토링 ON

멘토링 설정

- 홈
- 강의
- 로드맵
- 수강평
- 게시글
- 블로그

수정하기

이 되었던 때가 있었습니
"IT 엔지니어"라고 적고

을 가지고 있습니다. 누구
람입니다. 개발자로써, 강
지만, 그래도, 남들보다 조

tecture, Blockchain,

강의 전체 6

[개정판] 웹 애플리케이션 개발을 위한 IntelliJ IDEA 설정

학습중

0강 / 25강 (0%)

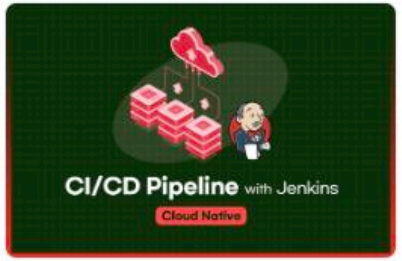
818명

멀티OS 사용을 위한 가상화 환경 구축 가이드 (Docker + Kubernetes)

학습중

0강 / 16강 (0%)

924명



Jenkins를 이용한 CI/CD Pipeline 구축

학습중

81강 / 83강 (98%)

독점 2709명

Spring Cloud로 개발하는 마이크로서비스 애플리케이션(MSA)

학습중

156강 / 156강 (100%)

독점 5531명

Spring Boot를 이용한 RESTful Web Services 개발

학습중

3강 / 48강 (6%)

독점 3862명

[구버전] 웹 애플리케이션 개발을 위한 IntelliJ IDEA 설정 (2020 ver.)

학습중

14강 / 14강 (100%)

4813명

목차

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Section 9.

Advanced Docker Usage

- Docker와 MSA
- 멀티 플랫폼 이미지를 위한 Docker Buildx
- Docker Runtime을 대신하는 기술
- Docker Registry를 대신하는 기술

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Cloud Native Architecture

- https://www.cncf.io/blog/2018/03/08/introducing-the-cloud-native-landscape-2-0-interactive-edition/

Microservices	Containerization	Continuous delivery	DevOps
A microservices architecture is an application development approach in which a large application is built as a suite of modular components or services	Containers are a type of software that can virtually package and isolate applications for deployment	Continuous delivery is a software delivery approach in which development teams produce and test code in short but continuous cycles	DevOps is a method-ology that promotes better communication and collaboration between development and operations teams

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Microservices

Microservices

A microservices architecture is an application development approach in which a large application is built as a suite of modular components or services

Containerization

Containers are a type of software that can virtually package and isolate applications for deployment

Continuous delivery

Continuous delivery is a software delivery approach in which development teams produce and test code in short but continuous cycles

DevOps

DevOps is a method-ology that promotes better communication and collaboration between development and operations teams

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Containerization

Microservices

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Continuous delivery

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DevOps

Microservices	Containerization	Continuous delivery	DevOps
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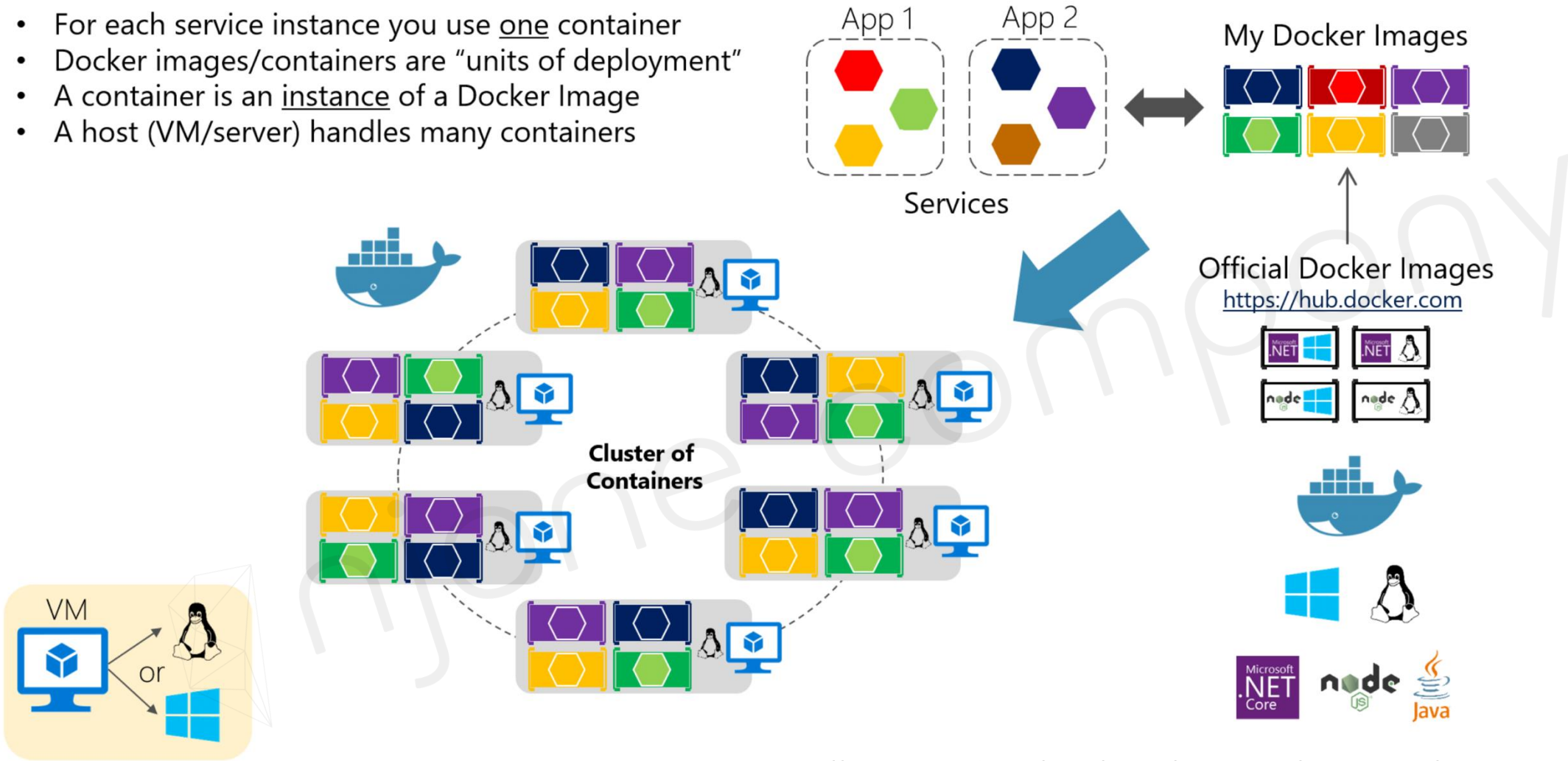
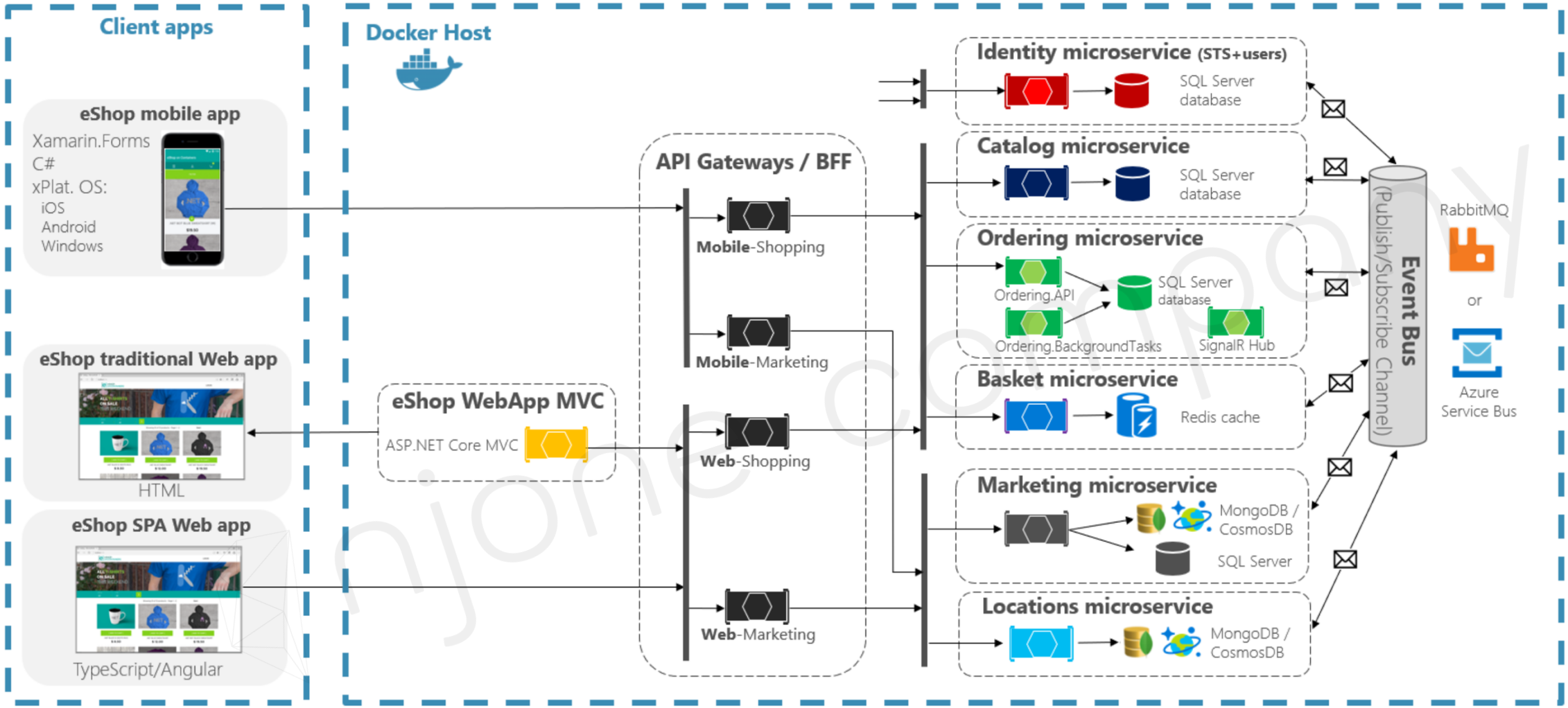
Composed Docker Applications in a Cluster

- For each service instance you use one container
- Docker images/containers are "units of deployment"
- A container is an instance of a Docker Image
- A host (VM/server) handles many containers



https://learn.microsoft.com/en-us/dotnet/architecture/microservices/architect-microservice-container-applications/scalable-available-multi-container-microservice-applications

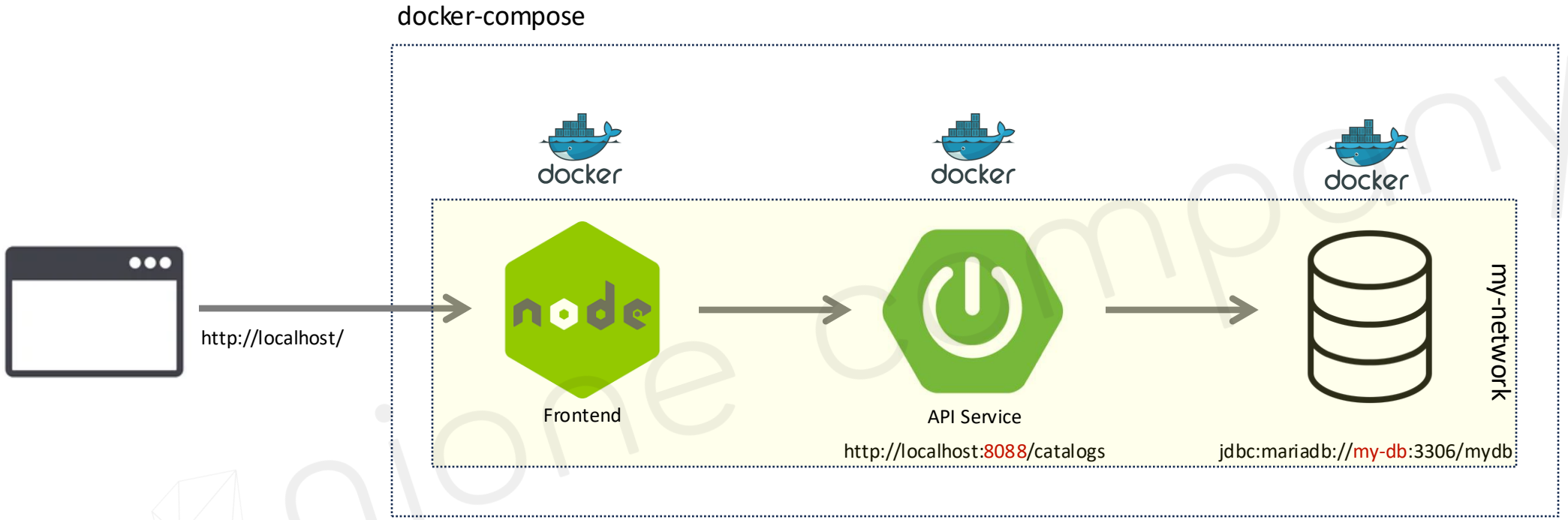
Development environment architecture



https://codeboulevard.com/2023/11/microservice-roadmap/

Docker와 MSA

MSA + Docker



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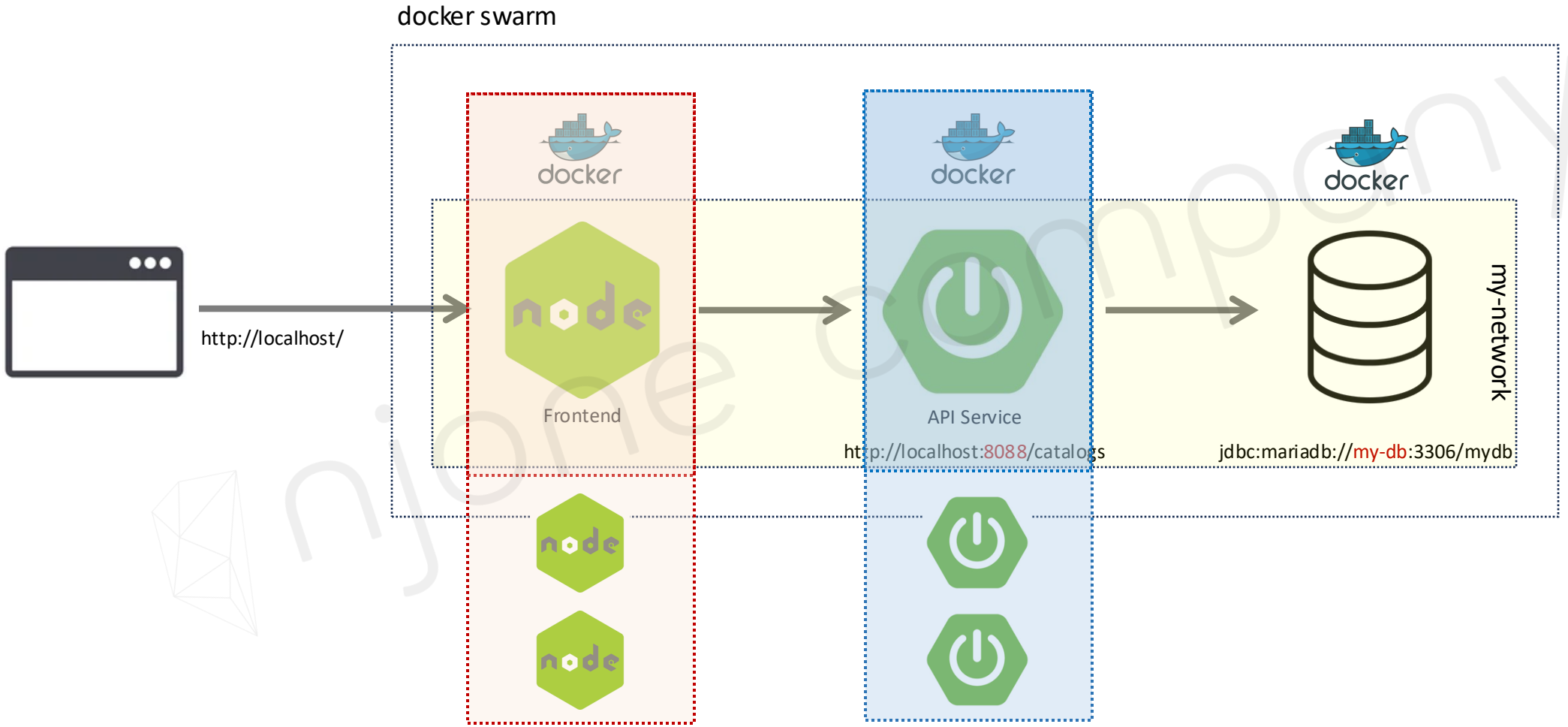
Docker와 MSA

MSA + Docker



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멀티 플랫폼 이미지를 위한 Docker Buildx

Docker Buildx?

- Multi Platform 빌드 기능 지원
  - Docker 19.03 부터 사용 가능
  - Docker Desktop → Buildx 플러그인 포함

- 지원 가능한 Platform

$ docker buildx ls

```
$ docker buildx ls
NAME/NODE           DRIVER/ENDPOINT     STATUS    BUILDKIT   PLATFORMS
default             docker
 \_ default          \_ default         running   v0.13.1    linux/arm64, linux/amd64, linux/amd64/v2, linux/riscv64,
 linux/ppc64le, linux/s390x, linux/386, linux/mips64le, linux/mips64, linux/arm/v7, linux/arm/v6
desktop-linux*      docker
 \_ desktop-linux    \_ desktop-linux   running   v0.13.1    linux/arm64, linux/amd64, linux/amd64/v2, linux/riscv64,
 linux/ppc64le, linux/s390x, linux/386, linux/mips64le, linux/mips64, linux/arm/v7, linux/arm/v6
```

```
$ docker buildx
Extended build capabilities with BuildKit

Usage:  docker buildx [OPTIONS] COMMAND

Extended build capabilities with BuildKit

Options:
      --builder string   Override the configured builder instance

Management Commands:
  imagetools  Commands to work on images in registry

Commands:
                                                    ce
                                                    to builder instance
                                                    tance
  rm          Remove one or more builder instances
  stop        Stop builder instance
  use         Set the current builder instance
  version     Show buildx version information
```

멀티 플랫폼 이미지를 위한 Docker Buildx

Cross-Platform Build

- MacOS Apple chip
  - Linux 이미지 빌드

```
FROM node:alpine
WORKDIR /app
```

```
$ docker build --tag edowon0623/my-node:m1 -f Dockerfile .
```

```
docker image inspect edowon0623/my-node:m1 | grep Architecture
        "Architecture": "arm64",
```

```
$ docker push edowon0623/my-node:m1
```

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Cross-Platform Build

- Linux 1)
  - Linux 이미지 사용

$ docker pull edowon0623/my-node:m1

```
[ec2-user@ip-172-31-5-111 ~]$ docker images | grep my-node
edowon0623/my-node          m1                  45b0e0a5bf15   6 minutes ago   137MB
```

$ docker run -it edowon0623/my-node:m1 bash

```
[ec2-user@ip-172-31-5-111 ~]$ docker run -it edowon0623/my-node:m1 bash
WARNING: The requested image's platform (linux/arm64/v8) does not match the detected host platform (linux/amd64) and
 no specific platform was requested
exec /usr/local/bin/docker-entrypoint.sh: exec format error
```

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멀티 플랫폼 이미지를 위한 Docker Buildx

Cross-Platform Build

- MacOS Apple chip
  - Linux 이미지 Buildx 빌드

$ docker buildx imagetools inspect node:alpine

```
docker buildx imagetools inspect node:alpine
Name:      docker.io/library/node:alpine
MediaType: application/vnd.docker.distribution.manifest.list.v2+json
Digest:    sha256:db8772d9f5796ac4e8c47508038c413ea1478da010568a2e48672f19a8b80cd2

Manifests:
  Name:      docker.io/library/node:alpine@sha256:c56ef510b1a041de46939462d89312015465895a6255434d012f1d2dcc8d85c7
  MediaType: application/vnd.docker.distribution.manifest.v2+json
  Platform:  linux/amd64

  Name:      docker.io/library/node:alpine@sha256:b84f96382d6a3b6b54add6172cf9627467e2b8b58e007471415773ed4d96c9c7
  MediaType: application/vnd.docker.distribution.manifest.v2+json
  Platform:  linux/arm/v6

  Name:      docker.io/library/node:alpine@sha256:58943268b3df5849ad18b8bfac735f8a30a35b4104a5269255a8adaacd168e72
  MediaType: application/vnd.docker.distribution.manifest.v2+json
  Platform:  linux/arm/v7

  Name:      docker.io/library/node:alpine@sha256:4233aea25ab7f7d288b23a0a80644da9fea41c4772ac4562840281e55c4ce54d
  MediaType: application/vnd.docker.distribution.manifest.v2+json
  Platform:  linux/arm64/v8
```

멀티 플랫폼 이미지를 위한 Docker Buildx

Cross-Platform Build

- MacOS Apple chip
  - Linux 이미지 Buildx 빌드

```
$ docker buildx build --platform linux/amd64 --tag edowon0623/my-node:amd63 -f Dockerfile .
```

```
> docker buildx build --platform linux/amd64 --tag edowon0623/my-node:amd64 -f Dockerfile .
[+] Building 1.0s (6/6) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                               0.0s
 => => transferring dockerfile: 66B                                                                0.0s
 => [internal] load metadata for docker.io/library/node:alpine                                     0.9s
 => [internal] load .dockerignore                                                                  0.0s
 => => transferring context: 2B                                                                    0.0s
 => [1/2] FROM docker.io/library/node:alpine@sha256:db8772d9f5796ac4e8c47508038c413ea1478da010568a2e48672f19a8  0.0s
 => CACHED [2/2] WORKDIR /app                                                                      0.0s
 => exporting to image                                                                             0.0s
 => => exporting layers                                                                            0.0s
 => => writing image sha256:8a3055aa4e6d6e658db532d343fa1d49a77d2e80020bc3cef75c8c894ee4972d       0.0s
 => => naming to docker.io/edowon0623/my-node:amd64                                                0.0s

Build multi-platform images faster with Docker Build Cloud: https://docs.docker.com/go/docker-build-cloud
```

멀티 플랫폼 이미지를 위한 Docker Buildx

Cross-Platform Build

- MacOS Apple chip
  - Linux 이미지 (amd64) 사용

```
> docker image inspect edowon0623/my-node:amd64 ¦ grep Architecture
        "Architecture": "amd64",
```

```
> docker run -it edowon0623/my-node:amd64 bash
WARNING: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and
no specific platform was requested
node:internal/modules/cjs/loader:1145
  throw err;
  ^

Error: Cannot find module '/app/bash'
    at Module._resolveFilename (node:internal/modules/cjs/loader:1142:15)
    at Module._load (node:internal/modules/cjs/loader:983:27)
    at Function.executeUserEntryPoint [as runMain] (node:internal/modules/run_main:142:12)
    at node:internal/main/run_main_module:28:49 {
  code: 'MODULE_NOT_FOUND',
  requireStack: []
}

Node.js v21.7.3
```

$ docker push edowon0623/my-node:amd64

Cross-Platform Build

- Linux 2)
  - Linux 이미지 사용

$ docker pull edowon0623/my-node:amd64

```
[ec2-user@ip-172-31-5-111 ~]$ docker images | grep my-node
edowon0623/my-node          amd64                8a3055aa4e6d   7 minutes ago    139MB
edowon0623/my-node          m1                   45b0e0a5bf15   20 minutes ago   137MB
```

$ docker run -it edowon0623/my-node:amd64 bash

```
[ec2-user@ip-172-31-5-111 ~]$ docker run -it edowon0623/my-node:amd64 sh
/app # node --version
v21.7.3
/app #
```

Docker Runtime을 대신하는 기술

CRI-O

- Docker의 한계점
  - Docker Server에 너무 많은 기능이 집중 → SPOF
  - Docker 명령어는 root 권한 필요

- CRI-O
  - Container Runtime Interface – Open Container Initiative
  - Container간 이식성 표준화를 위해 OCI 구성
  - Kubernetes 1.24 버전부터 Docker Runtime 제거 → Container Runtime 구성을 위해 CRI 사용
  - Container Runtime만 가능 → 이미지 빌드, CLI, Registry 기능 X
  - Container를 fork/exec. 방식으로 구동
  - Podman (Container 실행), Buildah (이미지 생성), Skopeo (이미지 저장소 관리)

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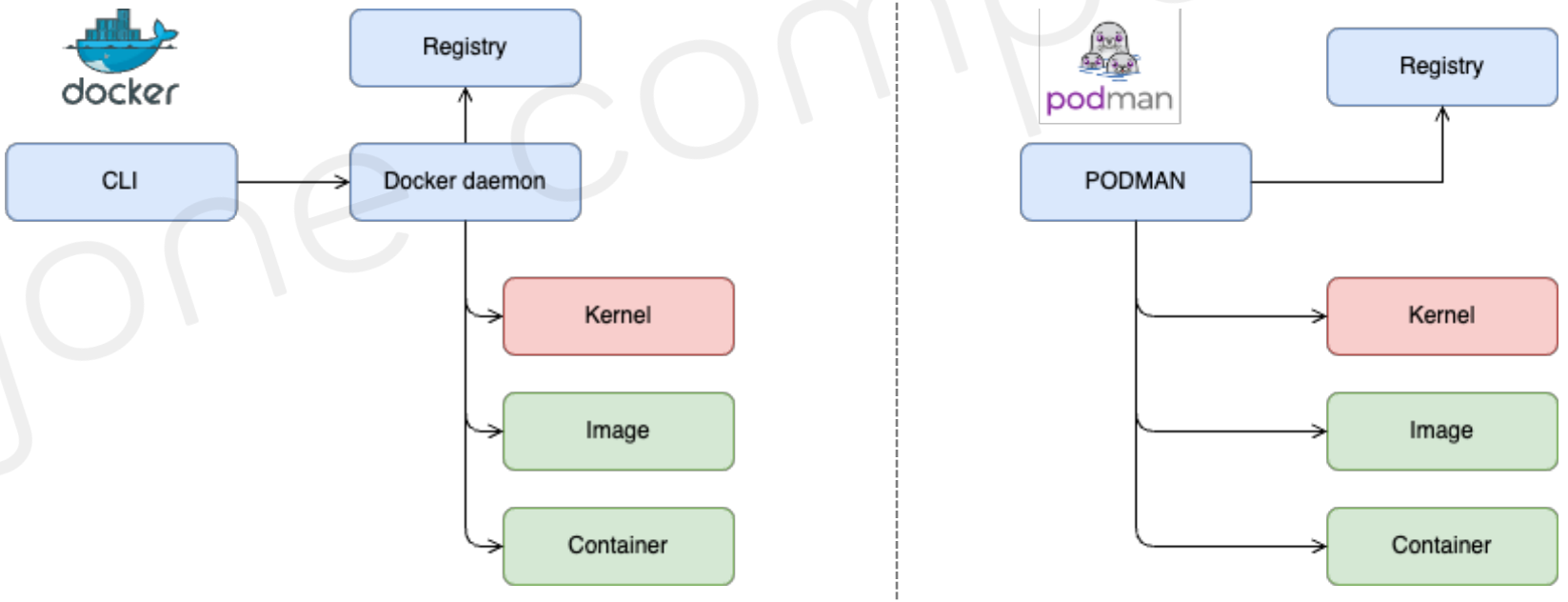
Docker Runtime을 대신하는 기술

PodMan

- Docker Cli와 같은 Tool
  - Container 생성,유지 및 관리 도구
  - Container를 fork/exec. 방식으로 구동 → daemon-less
  - root 권한 필요 X



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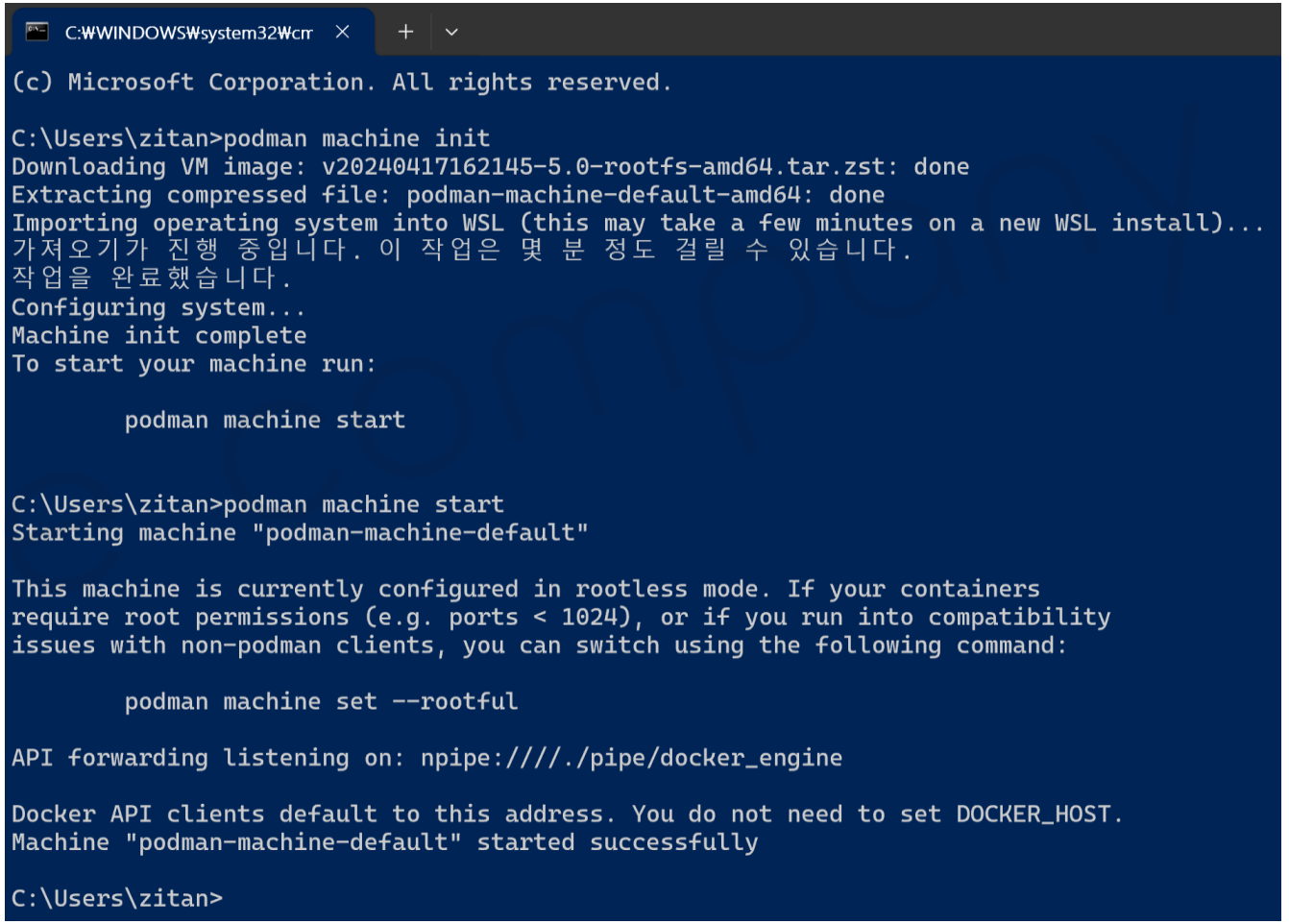
Docker Runtime을 대신하는 기술

Podman for Windows

- https://github.com/containers/podman/blob/main/docs/tutorials/podman-for-windows.md

```
C:\> podman machine init
C:\> podman machine start
C:\> podman info
```



Docker Runtime을 대신하는 기술

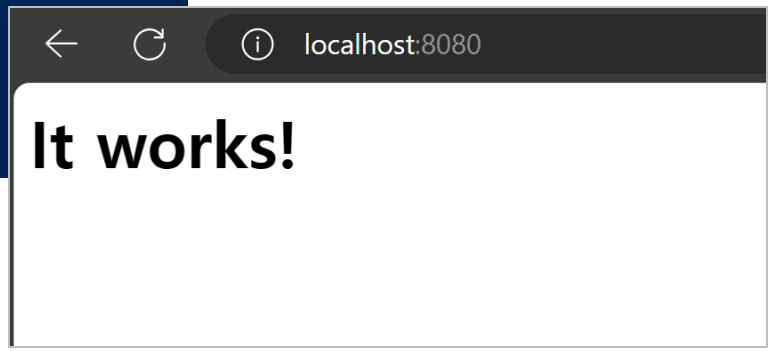
Podman for Windows

- Podman으로 Container 실행 - httpd

```
C:\> podman run --rm -d -p 8080:80 --name httpd docker.io/library/httpd
```

```
C:\Users\zitan>podman run --rm -d -p 8080:80 --name httpd docker.io/library/httpd
Trying to pull docker.io/library/httpd:latest...
Getting image source signatures
Copying blob sha256:d2b091e65160919e8aa45ccc4e2187a65be454ce437692ec811e3fe47386bd4f
Copying blob sha256:6e9a8835eae400bbb7b5572cc9aeea0a187e7216dd2def605c432cbe1d4955c5
Copying blob sha256:b927d001db705e0c4192a7c5ea2ca65c7c0f4858349d9a785e40e044d617724b
Copying blob sha256:13808c22b207b066ef43572e57e4fb8c6172e887dd9a918c089a174a19371b7a
Copying blob sha256:559cc51378ed1226b3d18ddb9d2149586a427338728bc7bd567bda03ca43b590
Copying blob sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1
Copying config sha256:fa0099f1c09d7d1dc852123c9456a23436df95715812df3db8b960ef6609e288
Writing manifest to image destination
1c63c0153fb1ff3a70abfb3b34a03bc6f6fdccd505d405665ca05403cf91751b

C:\Users\zitan>docker images
REPOSITORY   TAG       IMAGE ID       CREATED       SIZE
httpd        latest    fa0099f1c09d   2 weeks ago   152MB
```



Docker Runtime을 대신하는 기술

Podman for Windows

- Podman으로 Container 실행 - centos

C:\> podman run -it centos

```
C:\Users\zitan>podman run -it centos
Resolved "centos" as an alias (/etc/containers/registries.conf.d/000-shortnames.conf)
Trying to pull quay.io/centos/centos:latest...
Getting image source signatures
Copying blob sha256:7a0437f04f83f084b7ed68ad9c4a4947e12fc4e1b006b38129bac89114ec3621
Copying config sha256:300e315adb2f96afe5f0b2780b87f28ae95231fe3bdd1e16b9ba606307728f55
Writing manifest to image destination
[root@47e4bb5c8e3c /]#
```

```
C:\Users\zitan>podman ps -a
CONTAINER ID  IMAGE                           COMMAND           CREATED         STATUS                     PORTS                   NAMES
1c63c0153fb1  docker.io/library/httpd:latest  httpd-foreground  3 minutes ago   Up 3 minutes               0.0.0.0:8080->80/tcp    httpd
47e4bb5c8e3c  quay.io/centos/centos:latest    /bin/bash         49 seconds ago  Exited (127) 8 seconds ago                         jovial_mcclintock
```

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Docker Runtime을 대신하는 기술

Podman for MacOS (M1)

- https://podman.io/docs/installation

```
brew install qemu podman
==> Downloading https://formulae.brew.sh/api/formula.jws.json
######################################################################################################################## 100.0%
==> Downloading https://formulae.brew.sh/api/cask.jws.json
######################################################################################################################## 100.0%
==> Downloading https://ghcr.io/v2/homebrew/core/qemu/manifests/9.0.0
Already downloaded: /Users/edowon/Library/Caches/Homebrew/downloads/6d7f489eba08de6d463648d2ad0fcac872edb1733993437ad260c98a86329213-
==> Fetching qemu
```

```
==> podman
        In order to run containers locally, podman depends on a Linux kernel.
        One can be started manually using `podman machine` from this package.
        To start a podman VM automatically at login, also install the cask
        "podman-desktop".

zsh completions have been installed to:
  /usr/local/share/zsh/site-functions
(base)  edowon  ~/Desktop/Work/14.online/devops-docker  section5 ±
```

Docker Runtime을 대신하는 기술

Podman for MacOS (M1)

- https://podman.io/docs/installation

```
 podman machine init
Extracting compressed file: podman-machine-default_fedora-coreos-40.20240504.2.0-qemu.x86_64.qcow2: done
Image resized.
Machine init complete
To start your machine run:

        podman machine start

(base)  edowon  ~/Desktop/Work/14.online/devops-docker  section5 ±
```

```
 podman machine start
Starting machine "podman-machine-default"
Waiting for VM ...
Error: qemu exited unexpectedly with exit code -1, stderr: qemu-system-x86_64: Unknown Error
```

Docker Runtime을 대신하는 기술

Podman for MacOS (M3)

- https://podman.io/docs/installation

```
[redacted]-iMac ~ % brew install qemu podman
==> Downloading https://ghcr.io/v2/homebrew/core/qemu/manifests/9.0.0
##################################################################################################################
==> Fetching dependencies for qemu: capstone, dtc, pcre2, mpdecimal, ca-certificates, openssl@3, readline, sqlite, xz, pytho
p11-kit, libevent, libnghttp2, unbound, gnutls, jpeg-turbo, libpng, libslirp, libssh, libusb, lzo, ncurses, pixman, snappy,
==> Downloading https://ghcr.io/v2/homebrew/core/capstone/manifests/5.0.1
##################################################################################################################
==> Fetching capstone
==> Downloading https://ghcr.io/v2/homebrew/core/capstone/blobs/sha256:667f0e0d8960724b71c950ccc7769e747269b828c9e1c90009dc5
##################################################################################################################
==> Downloading https://ghcr.io/v2/homebrew/core/dtc/manifests/1.7.0
```

```
[redacted]-iMac ~ % podman machine init
Looking up Podman Machine image at quay.io/podman/machine-os:5.0 to create VM
Getting image source signatures
Copying blob 4532a1715d4d done   |
Copying config 44136fa355 done   |
Writing manifest to image destination
4532a1715d4d56948bae034da8c169e5fe0ee1739f8ded4268b836356e5f085d
Extracting compressed file: podman-machine-default-arm64.raw: done
Machine init complete
To start your machine run:

        podman machine start

naeun@ina-eun-ui-iMac ~ %
```

Docker Runtime을 대신하는 기술

Podman for MacOS (M3)

- https://podman.io/docs/installation

```
[redacted]-iMac ~ % podman machine start
Starting machine "podman-machine-default"

This machine is currently configured in rootless mode. If your containers
require root permissions (e.g. ports < 1024), or if you run into compatibility
issues with non-podman clients, you can switch using the following command:

        podman machine set --rootful

API forwarding listening on: /var/folders/fc/41cm3p451j51791m4sry7bpr0000gn/T/podman/podman-machine-default-api.sock

The system helper service is not installed; the default Docker API socket
address can't be used by podman. If you would like to install it, run the following commands:

        sudo /opt/homebrew/Cellar/podman/5.0.2/bin/podman-mac-helper install
        podman machine stop; podman machine start

You can still connect Docker API clients by setting DOCKER_HOST using the
following command in your terminal session:

        export DOCKER_HOST='unix:///var/folders/fc/41cm3p451j51791m4sry7bpr0000gn/T/podman/podman-machine-default-api.sock'

Machine "podman-machine-default" started successfully
```

```
[redacted]-iMac ~ % podman machine list
NAME                     VM TYPE     CREATED             LAST UP             CPUS        MEMORY      DISK SIZE
podman-machine-default*  applehv     About a minute ago  Currently running  4           2GiB        100GiB
naeun@ina-eun-ui-iMac ~ %
```

Docker Runtime을 대신하는 기술

Podman for MacOS (M3)

- https://podman.io/docs/installation

```
[redacted]-iMac ~ % podman run --rm -d -p 8080:80 --name httpd docker.io/library/httpd
Trying to pull docker.io/library/httpd:latest...
Getting image source signatures
Copying blob sha256:bae159c85a0b3662b8e9d58344f7da1b84836a7338c8ddf6b88360cea400929f
Copying blob sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1
Copying blob sha256:22d97f6a5d13532e867231d23d92620a81874d51a456196be50154eeb32edc08
Copying blob sha256:c521cc2c4e8a439951003122cdab841fe1b1350db864e2cf9dea69298513f711
Copying blob sha256:855b568ab02c3fee1362d51176f966517b1e03ca0c8efde55eddace90835f300
Copying blob sha256:98c8bced82bd1d3557a49d39ea03bb8090bf3abc8154ab65388d9f9728f3ce02
Copying config sha256:73acb239a8eb7a9ef5b6848a5b01bb26cc03d2262056705876bf7ac6ad781f02
Writing manifest to image destination
df1fe9a8ae626d75306733ac36609b8aa40fce2b5b2490b8c91a8a496fc941f5
naeun@ina-eun-ui-iMac ~ %
```

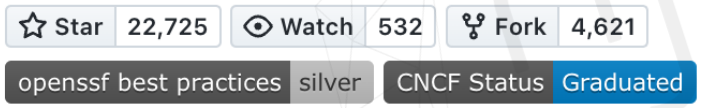
Docker Regitry를 대신하는 기술

Harbor

- https://goharbor.io/
- https://goharbor.io/docs/2.10.0/install-config/

What is Harbor?

Harbor is an open source registry that secures artifacts with policies and role-based access control, ensures images are scanned and free from vulnerabilities, and signs images as trusted. Harbor, a CNCF Graduated project, delivers compliance, performance, and interoperability to help you consistently and securely manage artifacts across cloud native compute platforms like Kubernetes and Docker.



Software	Version
Docker Engine	20.10.10-ce+ or higher
Docker Compose	1.18.0+
OpenSSL	Latest

Port	Protocol
443	HTTPS
4443	HTTPS
80	HTTP

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Docker Regitry를 대신하는 기술

인증서 설치

1) CA Certificates 생성
2) Server Certificates 생성
3) SAN(Subject Alternative Name) 등록
4) Docker Engine Certificate 업데이트

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Docker Regitry를 대신하는 기술

인증서 설치

1) CA Certificates 생성

- Root CA 비밀키 생성 → 공개키 생성

```
$ openssl genrsa –out ca.key 4096
```

```
$ openssl req -x509 -new -nodes -sha512 -days 365 -key ca.key -out ca.crt
```

2) Server Certificates 생성

- 서버의 인증서를 위한 비밀키 생성 → CSR(Certificate Signing Request) 파일 생성

```
$ openssl genrsa –out server.key 4096
```

```
$ openssl req -new -sha512 -key server.key -out server.csr
```

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인증서 설치

3) SAN(Subject Alternative Name) 등록

- CSR을 이용하여 SAN에 도메인 정보 추가 → 서버의 인증키 생성

```
$ vi v3ext.cnf
```

```
[root@45ad8a699362 certs]# cat v3ext.cnf
subjectAltName = IP:172.30.11.76,IP:127.0.0.1
```

Host PC IP address

```
$ openssl x509 -req -sha512 -days 365 \
    -extfile v3ext.cnf \
    -CA ca.crt -CAkey ca.key -CAcreateserial \
    -in server.csr \
    -out server.crt
```

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인증서 설치

4) Docker Engine Certificate 업데이트

- Docker Engine에서 사용 될 cert 파일 생성

```
$ openssl x509 -inform PEM -in server.crt -out server.cert
```

- Docker Engine에 인증서 파일 등록

```
$ mkdir -p /etc/docker/certs.d/server
$ cp server.cert /etc/docker/certs.d/server/
$ cp server.key /etc/docker/certs.d/server/
$ cp ca.crt /etc/docker/certs.d/server/
```

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Docker Regitry를 대신하는 기술

Harbor 설치

1) Harbor package 다운로드

- https://github.com/goharbor/harbor/releases

```
$ wget https://github.com/goharbor/harbor/releases/download/v2.2.2/harbor-offline-installer-v2.2.2.tgz
```

```
$ tag harbor-offline-installer-v2.2.2.tgz
```

2) harbor.yml 파일 생성 → 설정

```
$ cp harbor.yml.tmp harbor.yml
```

```
$ vi harbor.yml
```

Host PC IP address

```
hostname: 172.30.11.76

http:
  port: 80

https:
  port: 443
  certificate: /etc/docker/certs.d/server/server.cert
  private_key: /etc/docker/certs.d/server/server.key
```

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Docker Regitry를 대신하는 기술

Harbor 설치

3) Deploy

```
$ ./prepare
$ ./install.sh
```

```
[root@45ad8a699362 harbor]# docker-compose ps
NAME                COMMAND                  SERVICE             STATUS              PORTS
harbor-core         "/harbor/entrypoint.…"   core                running (healthy)
harbor-db           "/docker-entrypoint.…"   postgresql          running (healthy)
harbor-jobservice   "/harbor/entrypoint.…"   jobservice          running (healthy)
harbor-log          "/bin/sh -c /usr/loc…"   log                 running (healthy)   127.0.0.1:1514->1051
harbor-portal       "nginx -g 'daemon of…"   portal              running (healthy)
nginx               "nginx -g 'daemon of…"   proxy               running (healthy)   0.0.0.0:80->8080/tcp
redis               "redis-server /etc/r…"   redis               running (healthy)
registry            "/home/harbor/entryp…"   registry            running (healthy)
registryctl         "/home/harbor/start.…"   registryctl         running (healthy)
```

Docker Regitry를 대신하는 기술

Harbor 사용

1) Project 생성
2) *Container#1]* docker login –u admin <Harbor IP address>
3) *Container#1]* docker push <IMAGE>
4) *Container#2]* docker login –u user1 <Harbor IP address>
5) *Container#2]* docker pull <IMAGE>

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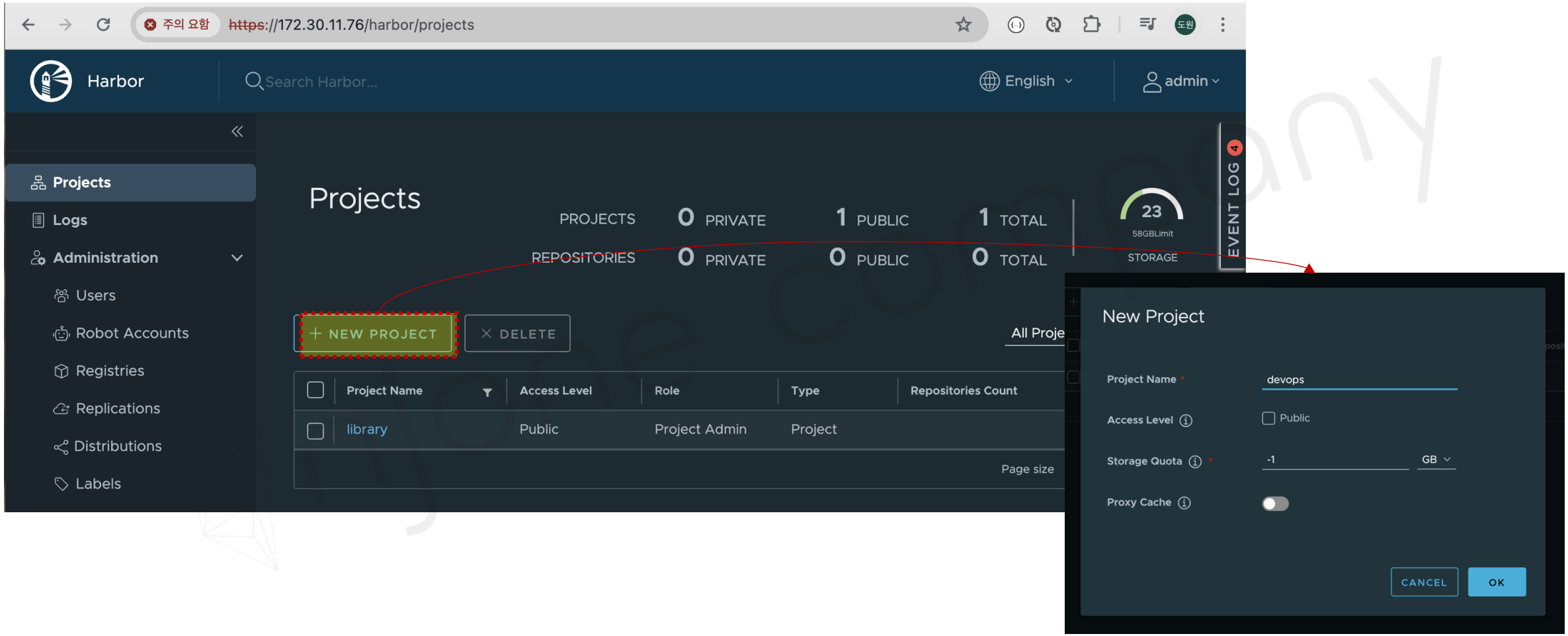
Docker Regitry를 대신하는 기술

Harbor 사용

1) Project 생성



Docker Regitry를 대신하는 기술

Harbor 사용

2) *Container#1]* docker login –u admin <Harbor IP address>

```
[root@45ad8a699362 harbor]# docker login -u admin 172.30.11.76
Password:
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[root@45ad8a699362 harbor]#
```

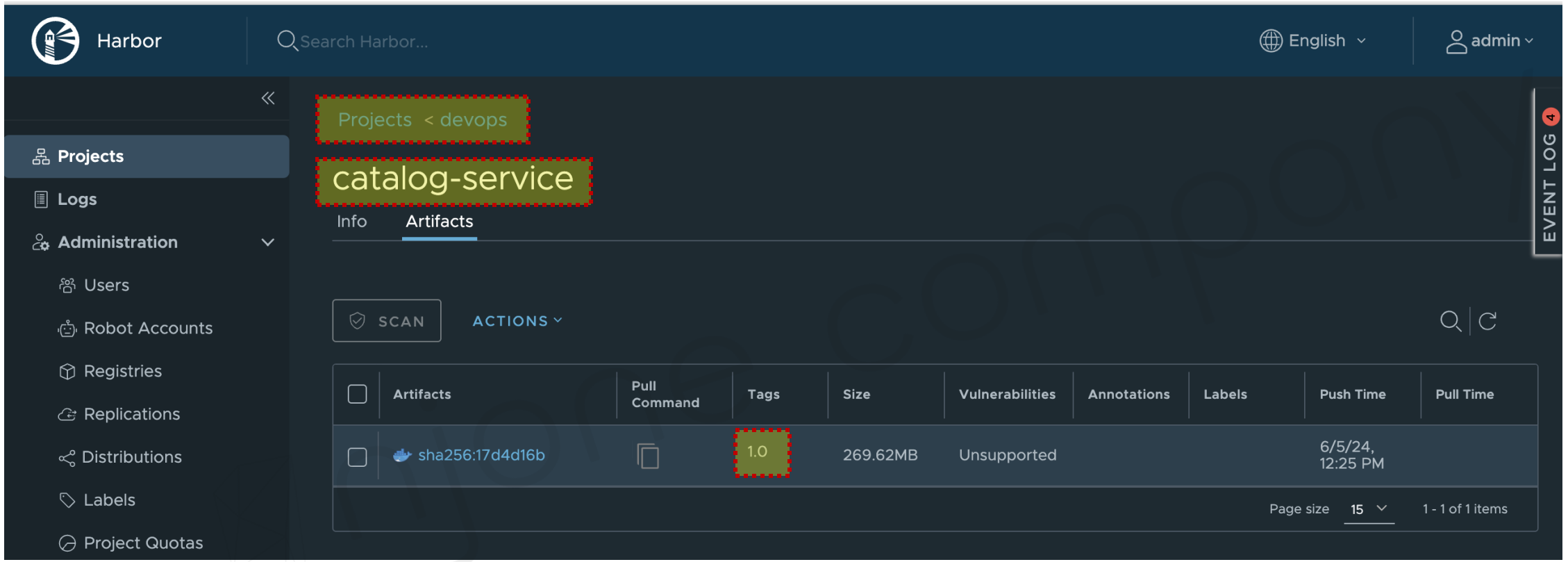
3) *Container#1]* docker push <IMAGE>

```
[root@45ad8a699362 harbor]# docker tag edowon0623/catalog-service:1.0 172.30.11.76/devops/catalog-service:1.0
[root@45ad8a699362 harbor]#
[root@45ad8a699362 harbor]# docker push 172.30.11.76/devops/catalog-service:1.0
The push refers to repository [172.30.11.76/devops/catalog-service]
b8029593adec: Layer already exists
3d3fdb9815af: Layer already exists
08664b16f94c: Layer already exists
9eb82f04c782: Layer already exists
1.0: digest: sha256:17d4d16bc218d5d686191edf72cc69a0d59b230d7a0c698dce915506d4ee30b5 size: 1165
```

Docker Regitry를 대신하는 기술

Harbor 사용

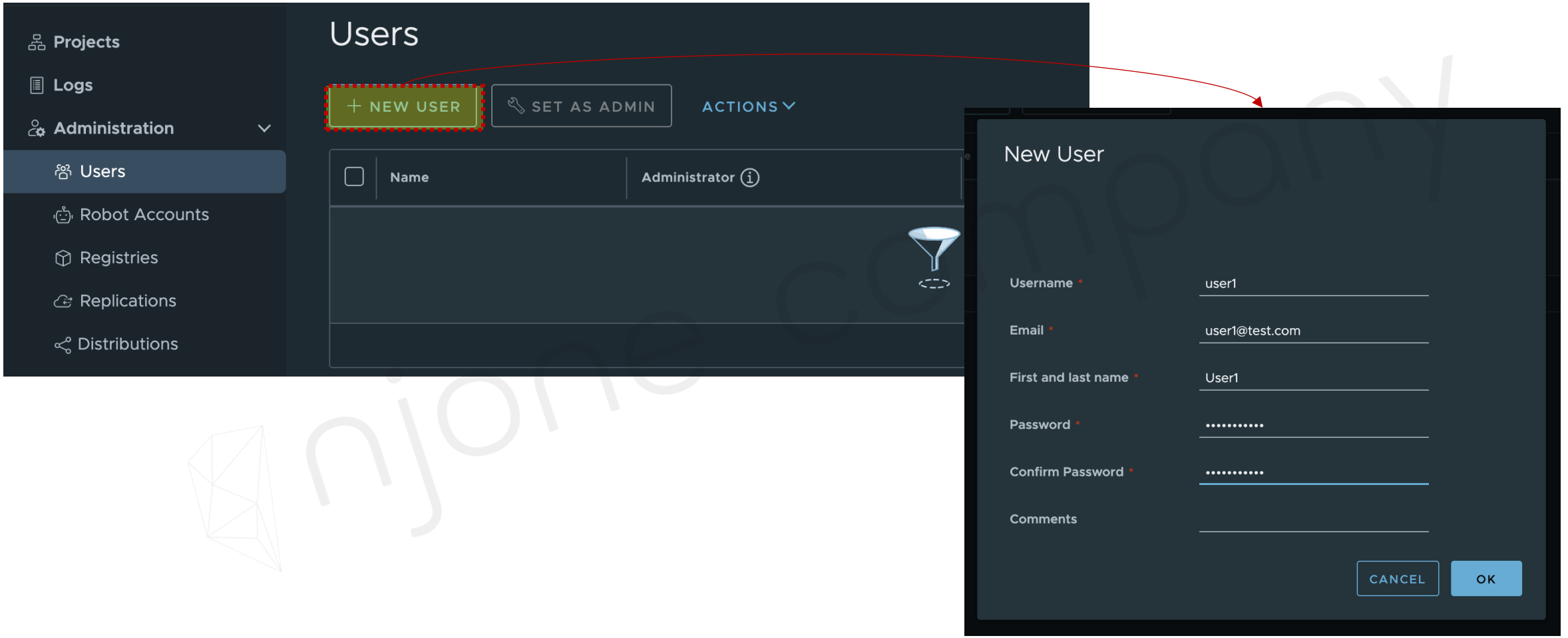
3) *Container#1]* docker push <IMAGE>



Docker Regitry를 대신하는 기술

Harbor 사용

4) *Container#2]* docker login –u user1 <Harbor IP address>



Docker Regitry를 대신하는 기술

Harbor 사용

4) *Container#2]* docker login –u user1 <Harbor IP address>

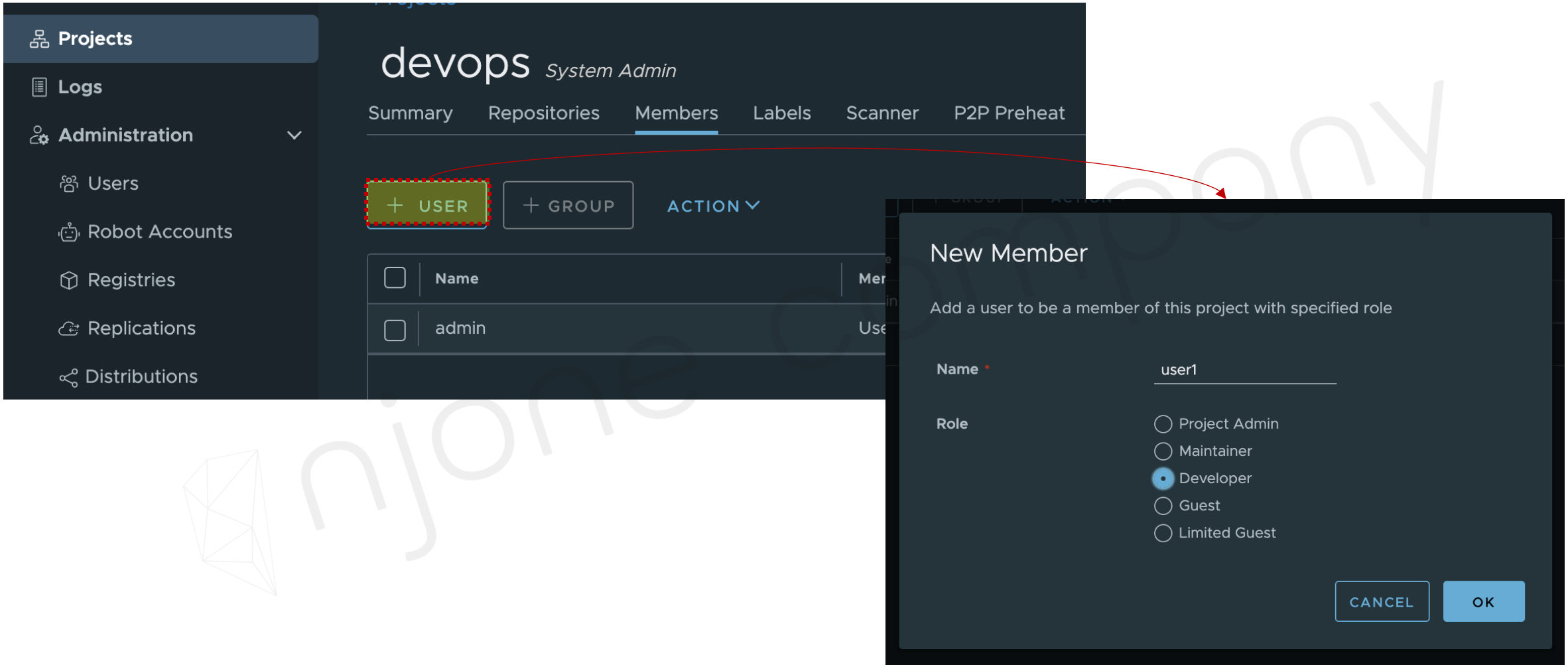
```
[root@33c30115eced ~]# docker login -u user1 172.30.11.76
Password:
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[root@33c30115eced ~]#
```

```
[root@33c30115eced ~]# docker pull 172.30.11.76/devops/catalog-service:1.0
Error response from daemon: unauthorized: unauthorized to access repository: devops/catalog-service,
action: pull: unauthorized to access repository: devops/catalog-service, action: pull
[root@33c30115eced ~]#
```

Harbor 사용

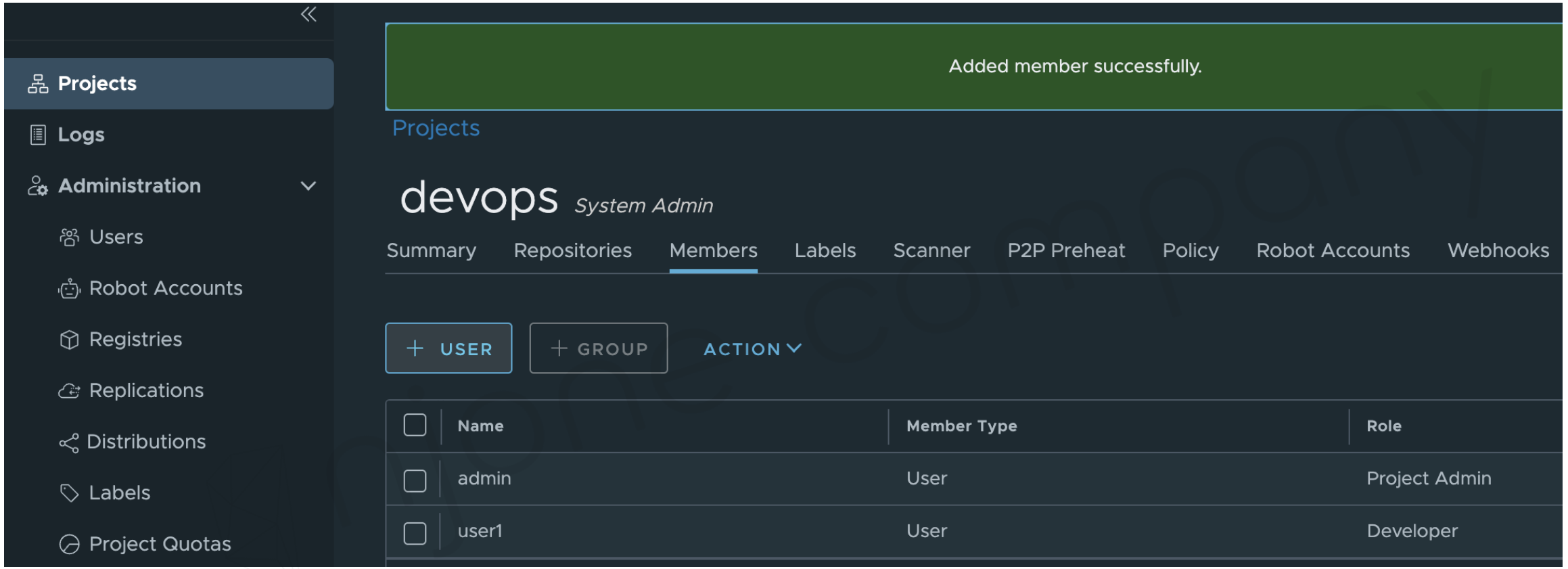
5) Container#2] docker pull <IMAGE>



Docker Regitry를 대신하는 기술

Harbor 사용

5) *Container#2]* docker pull <IMAGE>



Harbor 사용

5) *Container#2]* docker pull <IMAGE>

```
[root@33c30115eced ~]# docker pull 172.30.11.76/devops/catalog-service:1.0
1.0: Pulling from devops/catalog-service
45b42c59be33: Pull complete
a91c0c19c848: Pull complete
0a37071eae8e: Pull complete
17bec8782113: Pull complete
Digest: sha256:17d4d16bc218d5d686191edf72cc69a0d59b230d7a0c698dce915506d4ee30b5
Status: Downloaded newer image for 172.30.11.76/devops/catalog-service:1.0
172.30.11.76/devops/catalog-service:1.0
[root@33c30115eced ~]#
```

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